

Myke Vital Sao Temgoua

Department of Physics, Faculty of Science
University of Yaoundé I
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

September 10, 2026

Prof. Kenneth M. Merz, Jr., Editor-in-Chief
Journal of Chemical Information and Modeling
American Chemical Society

Dear Prof. Merz,

We submit the revised manuscript **“Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis”** for consideration as an Article in *Journal of Chemical Information and Modeling*.

This work addresses a methodological challenge in computational antimalarial discovery: evaluating target breadth and mutation resilience as distinct, complementary properties when prioritizing candidates from African natural-product-inspired chemical space, while maintaining clear boundaries between computational evidence and biological activity. A 17-member polypharmacology-oriented cohort was analyzed across four mechanistically diverse targets (PfDHFR, PfCRT, PfClpP, PfATP4) using target-specific structural anchors and a per-target resistance-resilience score (RRS). The workflow integrates three evidence layers: chemical-space expansion (65 856 hybrid-library molecules $\rightarrow$ 19 913 prioritized candidates), target-anchored AutoDock Vina docking generating a complete 17×4 scoring matrix (68/68 geometric quality pass), and per-target RRS computed separately for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels. PfCRT was re-docked against a corrected 3D7-like wild-type receptor under a cavity-anchored grid; a pipeline-null control showed no detectable K76T/K76A signal for this panel. The cohort yielded seven A*, four B, and six C RRS profiles (range 73.2 %–115.6 %). After multiplicity correction, PNS–RRS ($\rho = -0.714$), RRS–wild-type score ($\rho = 0.691$), and N_{fav} –RRS ($\rho = 0.714$) were significant; ACSI–RRS was not ($\rho = -0.190$).

Key elements of the revision:

1. **Protocol validation.** Three complementary checks: (a) external DEKOIS 2.0 benchmark (PfDHFR ROC-AUC 0.45, near-chance; controlled MTX-retained vs MTX-stripped comparison, 0.502 vs 0.563 with overlapping CIs, EF@1% zero in both arms); (b) MMV Malaria Box enrichment (ROC-AUC 0.924–1.000); (c) redocking of reference ligands (four of five reproduced with RMSD < 2.0 Å; the methotrexate exception reflects NADPH-adjacent grid placement).
2. **Retrospective boundary check.** RRS profiles for five approved antimalarials (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) against the same panels do not recover clinically documented resistance signatures (pyrimethamine ≈ 100 % on all four PfDHFR mutants; chloroquine ≈ 99 % on PfCRT K76T/K76A, both within the pipeline-null spread), reinforcing that docking-RRS values are not resistance-circumvention evidence.
3. **Physicochemical characterization.** All 17 candidates satisfy Lipinski/Veber criteria (100 % compliance; mean QED 0.703); the ADMET summary is population-level and is not a candidate-specific safety assessment.
4. **Methodological transparency.** Complete grid specifications, within-target median favorability, pipeline-null RRS control, multi-seed sensitivity, and formal definitions of PNS and ACSI are provided. Centroid-based library reduction yields a 99.3 % computational-cost reduction relative to exhaustive docking, with the caveat that this metric describes cost rather than active-compound retention.

Evidence boundaries. The study does not claim experimental affinity, confirmed simultaneous target engagement, biological resistance circumvention, or measured $\text{IC}_{50}/\text{EC}_{50}$ values. It produces (a) four-target docking profiles from which polypharmacology hypotheses can be generated, (b) per-target RRS values summarising mutation-panel docking ratios, and (c) exploratory cross-metric associations without conflation of evidence layers.

Reproducibility. Complete records (source code, candidate lists, receptor and ligand files, grid configurations, raw poses, RRS protocols, machine-readable tables) are archived in the versioned Zenodo deposit `zenodo_package_P1/` (sha256-verified, MIT license; DOI to be inserted upon assignment). The Supporting Information contains the validation datasets and methodological specifications.

The centroid-based reduction strategy lowers the computational barrier for research groups in endemic regions (94 % of malaria cases) by orders of magnitude while retaining structural diversity, offering a reproducible template for hypothesis generation in resistance-challenged therapeutic areas. This manuscript is original, is not under consideration elsewhere, and has been approved by all authors. We declare no competing financial interests.

Sincerely,

Myke Vital Sao Temgoua (on behalf of all co-authors)